Multicompartment distribution and dynamics of the antifungal resistome of *Candida tropicalis*, with emphasis on the azole-resistant clade 4: a systematic review and meta-analysis

SEARCH STRATEGY

Supplementary material to the study protocol

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1. SCOPE AND CONVENTIONS

This document reports the literal search strings executed in every source, with their yield and date, as required by PRISMA-S. It is supplementary to the study protocol and is to be read with it.

Boolean architecture. The master equation of each source is the intersection of the population block with the exposure block and with the union of the outcome and lineage blocks, in the form A and B and, in parentheses, D or C. Each of the four blocks is additionally executed as an independent line, in order to document its yield and to serve as a safety net for studies the intersection might miss.

Filters. No filter of language, publication date, document type, full-text availability or species was applied in any source.

Line of record. For each source, the line whose records enter screening is identified in section 3. In most sources this is the population line rather than the master equation, because a substantial proportion of records reporting the resistome do not name the compartment in indexed fields.

Dates and closure. Dates are those of actual execution. The search closure date is 12 September 2026, corresponding to the last source executed. Yields are not additive within or between sources; the consolidated total is obtained after import and deduplication.

Notation. Yields reported as two figures, such as 2,364 reported and 2,365 retrieved, distinguish the count the platform declares from the records actually obtained. Where a single figure is given, both coincide.

2. SEARCH STRINGS BY SOURCE

2.1 PubMed

Interface: pubmed.ncbi.nlm.nih.gov. Controlled vocabulary was verified in the search details of each line: all thirty-one descriptors resolved as MeSH terms and not as free text.

PM-A. Population block. Yield: 7,028; Date: 07/09/2026

"Candida tropicalis"[Mesh] OR "Candida tropicalis"[tiab] OR "C tropicalis"[tiab] OR "Candida paratropicalis"[tiab]

PM-B1. Compartment block, controlled vocabulary. Yield: 1,353,827; Date: 09/09/2026

"Candidiasis"[Mesh] OR "Candidemia"[Mesh] OR "Cross Infection"[Mesh] OR "Environmental Microbiology"[Mesh] OR "Soil Microbiology"[Mesh] OR "Water Microbiology"[Mesh] OR "Food Microbiology"[Mesh] OR "Animals"[Mesh] OR "Agriculture"[Mesh] OR "Fruit"[Mesh] OR "One Health"[Mesh]

PM-B2. Compartment block, free vocabulary. Yield: 3,412,703; Date: 09/09/2026

"clinical isolate"[tiab] OR "clinical isolates"[tiab] OR "isolation source"[tiab] OR colonization[tiab] OR colonisation[tiab] OR "hospital environment"[tiab] OR "healthcare environment"[tiab] OR nosocomial[tiab] OR fomite[tiab] OR fomites[tiab] OR "medical device"[tiab] OR "medical devices"[tiab] OR animal[tiab] OR animals[tiab] OR veterinary[tiab] OR livestock[tiab] OR poultry[tiab] OR swine[tiab] OR cattle[tiab] OR wildlife[tiab] OR agriculture[tiab] OR agricultural[tiab] OR agroecosystem[tiab] OR orchard[tiab] OR orchards[tiab] OR farm[tiab] OR farms[tiab] OR farming[tiab] OR crop[tiab] OR crops[tiab] OR fruit[tiab] OR fruits[tiab] OR rhizosphere[tiab] OR vineyard[tiab] OR plantation[tiab] OR greenhouse[tiab] OR postharvest[tiab] OR "irrigation water"[tiab] OR environmental[tiab] OR soil[tiab] OR groundwater[tiab] OR freshwater[tiab] OR seawater[tiab] OR sediment[tiab] OR sediments[tiab] OR sand[tiab] OR beach[tiab] OR estuary[tiab] OR estuarine[tiab] OR wetland[tiab] OR wetlands[tiab] OR mangrove[tiab] OR wastewater[tiab] OR "waste water"[tiab] OR sewage[tiab] OR effluent[tiab] OR sludge[tiab] OR "One Health"[tiab]

PM-B3. Agricultural azole pressure block. Yield: 27,722; Date: 09/09/2026

"Fungicides, Industrial"[Mesh] OR "azole fungicide"[tiab] OR "azole fungicides"[tiab] OR fungicide[tiab] OR fungicides[tiab] OR difenoconazole[tiab] OR tebuconazole[tiab] OR triadimenol[tiab] OR diniconazole[tiab] OR flutriafol[tiab] OR hexaconazole[tiab] OR propiconazole[tiab] OR epoxiconazole[tiab] OR prochloraz[tiab] OR imazalil[tiab] OR thiabendazole[tiab] OR "demethylation inhibitor"[tiab] OR "demethylation inhibitors"[tiab]

PM-B. Exposure and compartment block. Yield: 1,963; Date: 09/09/2026

("Candida tropicalis"[MeSH Terms] OR "Candida tropicalis"[Title/Abstract] OR "C tropicalis"[Title/Abstract] OR "Candida paratropicalis"[Title/Abstract]) AND ("Environmental Microbiology"[MeSH Terms] OR "Agriculture"[MeSH Terms] OR "crops, agricultural"[MeSH Terms] OR "Fruit"[MeSH Terms] OR "Soil"[MeSH Terms] OR "Seawater"[MeSH Terms] OR "Fresh Water"[MeSH Terms] OR "Wetlands"[MeSH Terms] OR "Ecosystem"[MeSH Terms] OR "Disease Reservoirs"[MeSH Terms] OR "animals, domestic"[MeSH Terms] OR "Livestock"[MeSH Terms] OR "Poultry"[MeSH Terms] OR "animals, wild"[MeSH Terms] OR "Veterinary Medicine"[MeSH Terms] OR "Cross Infection"[MeSH Terms] OR "Equipment Contamination"[MeSH Terms] OR "Fomites"[MeSH Terms] OR "Wastewater"[MeSH Terms] OR "One Health"[MeSH Terms] OR ("clinical isolate"[Title/Abstract] OR "clinical isolates"[Title/Abstract] OR "isolation source"[Title/Abstract] OR "colonization"[Title/Abstract] OR "colonisation"[Title/Abstract] OR "hospital environment"[Title/Abstract] OR "healthcare environment"[Title/Abstract] OR "nosocomial"[Title/Abstract] OR "fomite"[Title/Abstract] OR "Fomites"[Title/Abstract] OR "medical device"[Title/Abstract] OR "medical devices"[Title/Abstract] OR "animal"[Title/Abstract] OR "animals"[Title/Abstract] OR "veterinary"[Title/Abstract] OR "Livestock"[Title/Abstract] OR "Poultry"[Title/Abstract] OR "swine"[Title/Abstract] OR "cattle"[Title/Abstract] OR "wildlife"[Title/Abstract] OR "Agriculture"[Title/Abstract] OR "agricultural"[Title/Abstract] OR "agroecosystem"[Title/Abstract] OR "orchard"[Title/Abstract] OR "orchards"[Title/Abstract] OR "farm"[Title/Abstract] OR "farms"[Title/Abstract] OR "farming"[Title/Abstract] OR "crop"[Title/Abstract] OR "crops"[Title/Abstract] OR "Fruit"[Title/Abstract] OR "fruits"[Title/Abstract] OR "rhizosphere"[Title/Abstract] OR "vineyard"[Title/Abstract] OR "plantation"[Title/Abstract] OR "greenhouse"[Title/Abstract] OR "postharvest"[Title/Abstract] OR "irrigation water"[Title/Abstract] OR "environmental"[Title/Abstract] OR "Soil"[Title/Abstract] OR "groundwater"[Title/Abstract] OR "freshwater"[Title/Abstract] OR "Seawater"[Title/Abstract] OR "sediment"[Title/Abstract] OR "sediments"[Title/Abstract] OR "sand"[Title/Abstract] OR "beach"[Title/Abstract] OR "estuary"[Title/Abstract] OR "estuarine"[Title/Abstract] OR "wetland"[Title/Abstract] OR "Wetlands"[Title/Abstract] OR "mangrove"[Title/Abstract] OR "Wastewater"[Title/Abstract] OR "waste water"[Title/Abstract] OR "sewage"[Title/Abstract] OR "effluent"[Title/Abstract] OR "sludge"[Title/Abstract] OR "One Health"[Title/Abstract]) OR ("fungicides, industrial"[MeSH Terms] OR "azole fungicide"[Title/Abstract] OR "azole fungicides"[Title/Abstract] OR "fungicide"[Title/Abstract] OR "fungicides"[Title/Abstract] OR "difenoconazole"[Title/Abstract] OR "tebuconazole"[Title/Abstract] OR "triadimenol"[Title/Abstract] OR "diniconazole"[Title/Abstract] OR "flutriafol"[Title/Abstract] OR "hexaconazole"[Title/Abstract] OR "propiconazole"[Title/Abstract] OR "epoxiconazole"[Title/Abstract] OR "prochloraz"[Title/Abstract] OR "imazalil"[Title/Abstract] OR "thiabendazole"[Title/Abstract] OR "demethylation inhibitor"[Title/Abstract] OR "demethylation inhibitors"[Title/Abstract]))

PM-C1. Lineage and typing block, controlled vocabulary. Yield: 876,545; Date: 09/09/2026

"Molecular Typing"[Mesh] OR "Mycological Typing Techniques"[Mesh] OR "Multilocus Sequence Typing"[Mesh] OR "Whole Genome Sequencing"[Mesh] OR "Phylogeny"[Mesh] OR "Genotype"[Mesh] OR "Disease Transmission, Infectious"[Mesh]

PM-C2. Lineage and typing block, free vocabulary. Yield: 1,013,813; Date: 09/09/2026

clade[tiab] OR clades[tiab] OR "clade 4"[tiab] OR "clade IV"[tiab] OR "genotype 4"[tiab] OR "genotype IV"[tiab] OR "clonal complex"[tiab] OR genotype[tiab] OR genotypes[tiab] OR genotyping[tiab] OR "diploid sequence type"[tiab] OR DST[tiab] OR "multilocus sequence typing"[tiab] OR MLST[tiab] OR "molecular typing"[tiab] OR lineage[tiab] OR lineages[tiab] OR "population structure"[tiab] OR "genetic relatedness"[tiab] OR clonal[tiab] OR clonality[tiab] OR phylogenetic[tiab] OR phylogeny[tiab] OR phylogenomic[tiab] OR "whole genome sequencing"[tiab] OR WGS[tiab] OR "single nucleotide polymorphism"[tiab] OR SNP[tiab] OR microsatellite[tiab] OR "cross resistance"[tiab] OR "cross-transmission"[tiab] OR "zoonotic transmission"[tiab] OR "environmental transmission"[tiab] OR spillover[tiab]

PM-C. Lineage and typing block. Yield: 607; Date: 09/09/2026

("Candida tropicalis"[MeSH Terms] OR "Candida tropicalis"[Title/Abstract] OR "C tropicalis"[Title/Abstract] OR "Candida paratropicalis"[Title/Abstract]) AND ("Molecular Typing"[MeSH Terms] OR "Mycological Typing Techniques"[MeSH Terms] OR "Multilocus Sequence Typing"[MeSH Terms] OR "Whole Genome Sequencing"[MeSH Terms] OR "Phylogeny"[MeSH Terms] OR "Genotype"[MeSH Terms] OR "disease transmission, infectious"[MeSH Terms] OR ("clade"[Title/Abstract] OR "clades"[Title/Abstract] OR "clade 4"[Title/Abstract] OR "clade IV"[Title/Abstract] OR "genotype 4"[Title/Abstract] OR "genotype IV"[Title/Abstract] OR "clonal complex"[Title/Abstract] OR "Genotype"[Title/Abstract] OR "genotypes"[Title/Abstract] OR "genotyping"[Title/Abstract] OR "diploid sequence type"[Title/Abstract] OR "DST"[Title/Abstract] OR "Multilocus Sequence Typing"[Title/Abstract] OR "MLST"[Title/Abstract] OR "Molecular Typing"[Title/Abstract] OR "lineage"[Title/Abstract] OR "lineages"[Title/Abstract] OR "population structure"[Title/Abstract] OR "genetic relatedness"[Title/Abstract] OR "clonal"[Title/Abstract] OR "clonality"[Title/Abstract] OR "phylogenetic"[Title/Abstract] OR "Phylogeny"[Title/Abstract] OR "phylogenomic"[Title/Abstract] OR "Whole Genome Sequencing"[Title/Abstract] OR "WGS"[Title/Abstract] OR "single nucleotide polymorphism"[Title/Abstract] OR "SNP"[Title/Abstract] OR "microsatellite"[Title/Abstract] OR "cross resistance"[Title/Abstract] OR "cross-transmission"[Title/Abstract] OR "zoonotic transmission"[Title/Abstract] OR "environmental transmission"[Title/Abstract] OR "spillover"[Title/Abstract]))

PM-D1. Outcome and resistome block, controlled vocabulary. Yield: 241,771; Date: 09/09/2026

"Antifungal Agents"[Mesh] OR "Drug Resistance, Fungal"[Mesh] OR "Microbial Sensitivity Tests"[Mesh]

PM-D2. Outcome and resistome block, free vocabulary. Yield: 2,424,187; Date: 09/09/2026

resistance[tiab] OR resistant[tiab] OR susceptibility[tiab] OR susceptible[tiab] OR "reduced susceptibility"[tiab] OR "azole resistance"[tiab] OR "azole-resistant"[tiab] OR "fluconazole-resistant"[tiab] OR "azole tolerance"[tiab] OR resistome[tiab] OR "minimum inhibitory concentration"[tiab] OR MIC50[tiab] OR MIC90[tiab] OR "epidemiological cutoff"[tiab] OR "epidemiological cut-off"[tiab] OR ECV[tiab] OR ECOFF[tiab] OR "non-wild-type"[tiab] OR breakpoint[tiab] OR breakpoints[tiab] OR CLSI[tiab] OR EUCAST[tiab] OR "broth microdilution"[tiab] OR "disk diffusion"[tiab] OR "disc diffusion"[tiab] OR Etest[tiab] OR "E test"[tiab] OR fluconazole[tiab] OR voriconazole[tiab] OR itraconazole[tiab] OR posaconazole[tiab] OR isavuconazole[tiab] OR ketoconazole[tiab] OR clotrimazole[tiab] OR miconazole[tiab] OR "amphotericin B"[tiab] OR echinocandin[tiab] OR echinocandins[tiab] OR anidulafungin[tiab] OR caspofungin[tiab] OR micafungin[tiab] OR rezafungin[tiab] OR ibrexafungerp[tiab] OR flucytosine[tiab] OR fluorocytosine[tiab] OR "5-fluorocytosine"[tiab] OR "5-FC"[tiab] OR azole[tiab] OR azoles[tiab] OR triazole[tiab] OR triazoles[tiab] OR ERG11[tiab] OR ERG3[tiab] OR ERG5[tiab] OR Y132F[tiab] OR S154F[tiab] OR G464S[tiab] OR A395T[tiab] OR A395V[tiab] OR C461T[tiab] OR TAC1[tiab] OR MRR1[tiab] OR MRR2[tiab] OR UPC2[tiab] OR CDR1[tiab] OR CDR2[tiab] OR CDR3[tiab] OR MDR1[tiab] OR SNQ2[tiab] OR FKS1[tiab] OR FUR1[tiab] OR FCY1[tiab] OR FCY2[tiab] OR efflux[tiab] OR overexpression[tiab] OR "tandem duplication"[tiab] OR "copy number variation"[tiab] OR aneuploidy[tiab] OR isochromosome[tiab] OR "loss of heterozygosity"[tiab]

PM-D. Outcome and resistome block. Yield: 3,848; Date: 09/09/2026

("Candida tropicalis"[MeSH Terms] OR "Candida tropicalis"[Title/Abstract] OR "C tropicalis"[Title/Abstract] OR "Candida paratropicalis"[Title/Abstract]) AND ("Antifungal Agents"[MeSH Terms] OR "drug resistance, fungal"[MeSH Terms] OR "Microbial Sensitivity Tests"[MeSH Terms] OR ("resistance"[Title/Abstract] OR "resistant"[Title/Abstract] OR "susceptibility"[Title/Abstract] OR "susceptible"[Title/Abstract] OR "reduced susceptibility"[Title/Abstract] OR "azole resistance"[Title/Abstract] OR "azole-resistant"[Title/Abstract] OR "fluconazole-resistant"[Title/Abstract] OR "azole tolerance"[Title/Abstract] OR "resistome"[Title/Abstract] OR "minimum inhibitory concentration"[Title/Abstract] OR "MIC50"[Title/Abstract] OR "MIC90"[Title/Abstract] OR "epidemiological cutoff"[Title/Abstract] OR "epidemiological cut-off"[Title/Abstract] OR "ECV"[Title/Abstract] OR "ECOFF"[Title/Abstract] OR "non-wild-type"[Title/Abstract] OR "breakpoint"[Title/Abstract] OR "breakpoints"[Title/Abstract] OR "CLSI"[Title/Abstract] OR "EUCAST"[Title/Abstract] OR "broth microdilution"[Title/Abstract] OR "disk diffusion"[Title/Abstract] OR "disc diffusion"[Title/Abstract] OR "Etest"[Title/Abstract] OR "E test"[Title/Abstract] OR "fluconazole"[Title/Abstract] OR "voriconazole"[Title/Abstract] OR "itraconazole"[Title/Abstract] OR "posaconazole"[Title/Abstract] OR "isavuconazole"[Title/Abstract] OR "ketoconazole"[Title/Abstract] OR "clotrimazole"[Title/Abstract] OR "miconazole"[Title/Abstract] OR "amphotericin B"[Title/Abstract] OR "echinocandin"[Title/Abstract] OR "echinocandins"[Title/Abstract] OR "anidulafungin"[Title/Abstract] OR "caspofungin"[Title/Abstract] OR "micafungin"[Title/Abstract] OR "rezafungin"[Title/Abstract] OR "ibrexafungerp"[Title/Abstract] OR "flucytosine"[Title/Abstract] OR "fluorocytosine"[Title/Abstract] OR "5-fluorocytosine"[Title/Abstract] OR "5-FC"[Title/Abstract] OR "azole"[Title/Abstract] OR "azoles"[Title/Abstract] OR "triazole"[Title/Abstract]

OR "triazoles"[Title/Abstract] OR "ERG11"[Title/Abstract] OR "ERG3"[Title/Abstract] OR "ERG5"[Title/Abstract] OR "Y132F"[Title/Abstract] OR "S154F"[Title/Abstract] OR "G464S"[Title/Abstract] OR "A395T"[Title/Abstract] OR "A395V"[Title/Abstract] OR "C461T"[Title/Abstract] OR "TAC1"[Title/Abstract] OR "MRR1"[Title/Abstract] OR "MRR2"[Title/Abstract] OR "UPC2"[Title/Abstract] OR "CDR1"[Title/Abstract] OR "CDR2"[Title/Abstract] OR "CDR3"[Title/Abstract] OR "MDR1"[Title/Abstract] OR "SNQ2"[Title/Abstract] OR "FKS1"[Title/Abstract] OR "FUR1"[Title/Abstract] OR "FCY1"[Title/Abstract] OR "FCY2"[Title/Abstract] OR "efflux"[Title/Abstract] OR "overexpression"[Title/Abstract] OR "tandem duplication"[Title/Abstract] OR "copy number variation"[Title/Abstract] OR "aneuploidy"[Title/Abstract] OR "isochromosome"[Title/Abstract] OR "loss of heterozygosity"[Title/Abstract]))

PM-M. Master equation, intersection architecture. Yield: 1,194; Date: 09/09/2026

(PM-A) AND (PM-B1 OR PM-B2 OR PM-B3) AND (PM-C1 OR PM-C2 OR PM-D1 OR PM-D2)

The master equation is executed through the numeric history references of the platform. Composed strings using the hash-letter syntax of the block labels are not valid in this interface, a limitation verified and recorded.

2.2 Scopus

Interface: scopus.com. The keyword field is retained within the TITLE-ABS-KEY operator: restriction to title and abstract alone would have reduced the population line from 14,500 to 3,698 records, with 3,559 exclusive records lost, corresponding to articles in which the species or the resistance terms appear only in the indexing, a profile characteristic of studies from non-clinical compartments.

SC-A. Population block. Yield: 14,500; Date: 08/09/2026

TITLE-ABS-KEY("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis")

SC-B. Compartment block. Yield: 17,458,781; Date: 08/09/2026

TITLE-ABS-KEY("clinical isolate*" OR "isolation source" OR colonization OR colonisation OR "hospital environment" OR nosocomial OR fomite* OR "medical device*" OR animal* OR veterinary OR livestock OR poultry OR swine OR cattle OR wildlife OR agricultur* OR agroecosystem OR orchard* OR farm* OR crop* OR fruit* OR rhizosphere OR "irrigation water" OR environmental OR soil OR groundwater OR freshwater OR seawater OR sediment OR sand OR estuar* OR wetland* OR mangrove OR wastewater OR sewage OR effluent OR sludge OR "One Health")

SC-C. Agricultural azole pressure block. Yield: 67,368; Date: 08/09/2026

TITLE-ABS-KEY("azole fungicide*" OR fungicide* OR tebuconazole OR difenoconazole OR propiconazole OR epoxiconazole OR prochloraz OR imazalil OR thiabendazole OR triadimenol OR diniconazole OR flutriafol OR hexaconazole)

SC-D. Lineage and typing block. Yield: 2,044,767; Date: 08/09/2026

TITLE-ABS-KEY(clade* OR "clade 4" OR "clade IV" OR genotype* OR genotyping OR "diploid sequence type" OR DST OR "multilocus sequence typing" OR MLST OR "molecular typing" OR lineage OR "population structure" OR "clonal complex" OR "genetic relatedness" OR clonal* OR phylogen* OR "whole genome sequencing" OR WGS OR "single nucleotide polymorphism" OR microsatellite)

SC-E. Outcome and resistome block. Yield: 5,462,189; Date: 08/09/2026

TITLE-ABS-KEY(resistan* OR susceptib* OR "antifungal resistance" OR "azole resistance" OR "fluconazole resistance" OR "azole tolerance" OR resistome OR "multidrug resistance" OR "minimum inhibitory concentration" OR "epidemiological cutoff*" OR ECV OR ECOFF OR "non-wild-type" OR breakpoint* OR CLSI OR EUCAST OR "broth microdilution" OR "disk diffusion" OR fluconazole OR voriconazole OR itraconazole OR posaconazole OR isavuconazole OR ketoconazole OR clotrimazole OR miconazole OR "amphotericin B" OR echinocandin* OR anidulafungin OR caspofungin OR micafungin OR flucytosine OR "5-fluorocytosine" OR azole* OR triazole* OR ERG11 OR ERG3 OR ERG5 OR Y132F OR S154F OR A395T OR C461T OR TAC1 OR MRR1 OR UPC2 OR CDR1 OR CDR2 OR MDR1 OR FKS1 OR FUR1 OR efflux OR overexpression OR aneuploidy OR isochromosome OR "loss of heterozygosity")

SC-M. Master equation, intersection architecture. Yield: 2,715; Date: 08/09/2026

((TITLE-ABS-KEY ("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis")) AND ((TITLE-ABS-KEY ("clinical isolate" OR "isolation source" OR colonization OR colonisation OR "hospital environment" OR "healthcare environment" OR nosocomial OR fomite OR "medical device" OR animal OR veterinary OR livestock OR poultry OR swine OR cattle OR wildlife OR agriculture OR agricultural OR

agroecosystem OR orchard OR farm OR farming OR crop OR fruit OR rhizosphere OR vineyard OR plantation OR greenhouse OR postharvest OR "irrigation water" OR environmental OR soil OR groundwater OR freshwater OR seawater OR sediment OR sand OR beach OR estuary OR estuarine OR wetland OR mangrove OR "environmental reservoir" OR wastewater OR "waste water" OR sewage OR effluent OR sludge OR "One Health")) OR (TITLE-ABS-KEY ("azole fungicide" OR fungicide OR difenoconazole OR tebuconazole OR triadimenol OR diniconazole OR flutriafol OR hexaconazole OR propiconazole OR epoxiconazole OR prochloraz OR imazalil OR thiabendazole OR "demethylation inhibitor" OR "DMI fungicide")))) AND ((TITLE-ABS-KEY (resistan* OR susceptib* OR "azole tolerance" OR resistome OR "minimum inhibitory concentration" OR MIC50 OR MIC90 OR "epidemiological cutoff" OR ECV OR ECOFF OR "non-wild-type" OR breakpoint OR CLSI OR EUCAST OR "broth microdilution" OR "disk diffusion" OR "disc diffusion" OR Etest OR "E-test" OR fluconazole OR voriconazole OR itraconazole OR posaconazole OR isavuconazole OR ketoconazole OR clotrimazole OR miconazole OR "amphotericin B" OR echinocandin OR anidulafungin OR caspofungin OR micafungin OR rezafungin OR ibrexafungerp OR flucytosine OR fluorocytosine OR "5-fluorocytosine" OR "5-FC" OR azole OR triazole OR ERG11 OR ERG3 OR ERG5 OR Y132F OR S154F OR G464S OR A395T OR A395V OR C461T OR TAC1 OR MRR1 OR MRR2 OR UPC2 OR CDR1 OR CDR2 OR CDR3 OR MDR1 OR SNQ2 OR FKS1 OR FUR1 OR FCY1 OR FCY2 OR efflux OR overexpression OR "tandem duplication" OR "copy number variation" OR aneuploidy OR isochromosome OR "loss of heterozygosity")) OR (TITLE-ABS-KEY (clade OR "clade 4" OR "clade IV" OR "genotype 4" OR "genotype IV" OR "clonal complex" OR genotype OR genotyping OR "diploid sequence type" OR DST OR "multilocus sequence typing" OR MLST OR "molecular typing" OR lineage OR "population structure" OR "genetic relatedness" OR clonal OR clonality OR phylogen* OR "whole genome sequencing" OR WGS OR "single nucleotide polymorphism" OR SNP OR microsatellite OR "cross resistance" OR "cross transmission" OR "zoonotic transmission" OR "environmental transmission" OR spillover)))

2.3 ScienceDirect

Interface: sciencedirect.com. The platform caps each query at eight boolean operators and does not allow combination of searches, so no master equation is possible and the strategy is decomposed into fifteen sweeps. Quotation marks reduce recall on this platform, so the strategy operates without them except for the species name. Yields are not additive: the sum of the fifteen sweeps is 1,149 records, which exceeds the real number because of overlap between sweeps.

SD-1. Phenotypic resistance. Yield: 187; Date: 09/09/2026

"Candida tropicalis" AND (antifungal resistance OR antifungal susceptibility OR azole resistance OR drug resistance)

SD-2. Criteria and methodology. Yield: 172; Date: 09/09/2026

"Candida tropicalis" AND (minimum inhibitory concentration OR broth microdilution OR CLSI OR EUCAST)

SD-3. Azoles. Yield: 240; Date: 09/09/2026

"Candida tropicalis" AND (fluconazole OR voriconazole OR itraconazole OR posaconazole)

SD-4. Polyenes and echinocandins. Yield: 159; Date: 09/09/2026

"Candida tropicalis" AND (amphotericin B OR anidulafungin OR caspofungin OR micafungin)

SD-5. Target determinants. Yield: 25; Date: 09/09/2026

"Candida tropicalis" AND (ERG11 OR ERG3 OR Y132F OR S154F)

SD-6. Transporters and regulators. Yield: 15; Date: 09/09/2026

"Candida tropicalis" AND (CDR1 OR MDR1 OR TAC1 OR UPC2)

SD-7. Agricultural compartment. Yield: 33; Date: 09/09/2026

"Candida tropicalis" AND (agricultural OR orchard OR crop OR postharvest)

SD-8. Agricultural and environmental matrices. Yield: 55; Date: 09/09/2026

"Candida tropicalis" AND (soil OR fruit OR sediment OR seawater)

SD-9. Animal compartment. Yield: 6; Date: 09/09/2026

"Candida tropicalis" AND (animal OR veterinary OR livestock OR poultry)

SD-10. Healthcare compartment. Yield: 84; Date: 09/09/2026

"Candida tropicalis" AND (hospital environment OR fomite OR medical device OR colonization)

SD-11. Azole fungicides. Yield: 8; Date: 09/09/2026

"Candida tropicalis" AND (tebuconazole OR difenoconazole OR propiconazole OR azole fungicide)

SD-12. Typing and lineage. Yield: 33; Date: 09/09/2026

"Candida tropicalis" AND (clade OR MLST OR genotyping OR multilocus sequence typing)

SD-13. Genomics and genetic relatedness. Yield: 65; Date: 09/09/2026

"Candida tropicalis" AND (whole genome sequencing OR phylogenetic OR cross-resistance OR transmission)

SD-14. Anthropogenic matrix. Yield: 47; Date: 09/09/2026

"Candida tropicalis" AND (wastewater OR sewage OR effluent OR beach)

SD-15. Echinocandins and flucytosine, determinants. Yield: 20; Date: 09/09/2026

"Candida tropicalis" AND (FKS1 OR flucytosine OR FUR1 OR aneuploidy)

2.4 LILACS, Virtual Health Library portal

Interface: lilacs.bvsalud.org. The complete LILACS Plus collection was retained rather than filtering strictly to LILACS, since filtering would have discarded VETINDEX, a regional veterinary literature database pertinent to the animal compartment, together with the Argentine, Cuban, Paraguayan and Peruvian national databases. Of the 1,857 records returned by the line of record, 1,668 are unique: the portal returns the same record more than once when it is indexed in several of the databases integrating the collection.

LIL-1. Population, field search with descriptor. Yield: 1,854; Date: 10/09/2026

ti:("Candida tropicalis") OR ab:("Candida tropicalis") OR mh:("Candida tropicalis") OR ti:("C. tropicalis") OR ab:("C. tropicalis")

LIL-2. Population, all words. Yield: 1,857 returned, 1,668 unique; Date: 10/09/2026

tw:("Candida tropicalis") OR tw:("C. tropicalis") OR tw:("Candida paratropicalis")

LIL-3. Yeasts in non-clinical matrices with resistome. Yield: 388; Date: 10/09/2026

(tw:(levaduras OR leveduras OR yeasts)) AND (tw:(suelo OR solo OR fruta OR frutas OR agrícola OR huerto OR pomar OR cultivo OR sedimento OR arena OR areia OR veterinario OR veterinária OR aves OR porcino OR suíno OR bovino OR silvestre)) AND (tw:(resistencia OR resistência OR antifúngico OR antifúngica OR antifungal OR fluconazol OR sensibilidad OR sensibilidade))

LIL-4. Population and resistome in regional vocabulary. Yield: 1,144; Date: 10/09/2026

(tw:("Candida tropicalis") OR tw:("C. tropicalis")) AND (tw:(resistencia OR resistência OR resistente OR antifúngico OR antifúngica OR antifúngicos OR antifungal OR fluconazol OR fluconazole OR voriconazol OR itraconazol OR anfotericina OR sensibilidad OR sensibilidade OR susceptibilidad OR "concentración inhibitoria mínima" OR CIM OR CLSI OR EUCAST OR ERG11 OR clado OR clade OR genotipificación OR MLST))

Verification: LIL-2 contains LIL-1 in full, with zero records exclusive to the latter and a difference of three records.

2.5 Dimensions

Interface: app.dimensions.ai, open public version. The search is executed with the field selector set to title and abstract, not to full data. A differential test established that the same population line yields 11,364 records in title and abstract and 89,166 in full data, a factor of 7.8, because the full data scope includes the whole text of documents when the platform holds it.

DIM-A. Population block. Yield: 11,364; Date: 10/09/2026

"Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis"

DIM-B. Exposure and compartment block. Yield: 3,134; Date: 10/09/2026

("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") AND ("clinical isolate" OR "isolation source" OR colonization OR colonisation OR "hospital environment" OR nosocomial OR fomite OR "medical device" OR animal OR animals OR veterinary OR livestock OR poultry OR swine OR cattle OR wildlife OR agriculture OR agricultural OR agroecosystem OR orchard OR orchards OR farm OR farms OR crop OR crops OR fruit OR fruits OR rhizosphere OR vineyard OR greenhouse OR postharvest OR "irrigation water" OR environmental OR soil OR groundwater OR freshwater OR seawater OR sediment OR sand OR beach OR estuarine OR wetland OR mangrove OR wastewater OR sewage OR effluent OR sludge OR "One Health" OR "azole fungicide" OR fungicide OR fungicides OR tebuconazole OR difenoconazole OR propiconazole OR prochloraz OR imazalil OR thiabendazole)

DIM-C. Lineage and typing block. Yield: 512; Date: 10/09/2026

("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") AND (clade OR clades OR "clade 4" OR "clade IV" OR "genotype 4" OR "clonal complex" OR genotype OR genotyping OR "diploid sequence type" OR "multilocus sequence typing" OR MLST OR "molecular typing" OR lineage OR "population structure" OR "genetic relatedness" OR clonal OR phylogenetic OR phylogenomic OR "whole genome sequencing" OR WGS OR "single nucleotide polymorphism" OR microsatellite OR "cross resistance" OR "zoonotic transmission" OR "environmental transmission" OR spillover)

DIM-D. Outcome and resistome block. Yield: 5,182; Date: 10/09/2026

("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") AND (resistance OR resistant OR susceptibility OR susceptible OR "azole tolerance" OR resistome OR "minimum inhibitory concentration" OR "epidemiological cutoff" OR ECOFF OR "non-wild-type" OR breakpoint OR breakpoints OR CLSI OR EUCAST OR "broth microdilution" OR "disk diffusion" OR fluconazole OR voriconazole OR itraconazole OR posaconazole OR isavuconazole OR ketoconazole OR clotrimazole OR miconazole OR "amphotericin B" OR echinocandin OR echinocandins OR anidulafungin OR caspofungin OR micafungin OR flucytosine OR "5-fluorocytosine" OR azole OR azoles OR triazole OR triazoles OR ERG11 OR ERG3 OR ERG5 OR Y132F OR S154F OR A395T OR C461T OR TAC1 OR MRR1 OR UPC2 OR CDR1 OR CDR2 OR MDR1 OR FKS1 OR FUR1 OR efflux OR overexpression OR aneuploidy OR isochromosome OR "loss of heterozygosity")

DIM-M. Master equation, intersection architecture. Yield: 1,555; Date: 10/09/2026

("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") AND ("clinical isolate" OR "isolation source" OR colonization OR colonisation OR "hospital environment" OR nosocomial OR fomite OR "medical device" OR animal OR animals OR veterinary OR livestock OR poultry OR swine OR cattle OR wildlife OR agriculture OR agricultural OR agroecosystem OR orchard OR orchards OR farm OR farms OR crop OR crops OR fruit OR fruits OR rhizosphere OR vineyard OR greenhouse OR postharvest OR "irrigation water" OR environmental OR soil OR groundwater OR freshwater OR seawater OR sediment OR sand OR beach OR estuarine OR wetland OR mangrove OR wastewater OR sewage OR effluent OR sludge OR "One Health" OR "azole fungicide" OR fungicide OR fungicides OR tebuconazole OR difenoconazole OR propiconazole OR prochloraz OR imazalil OR thiabendazole) AND (clade OR clades OR "clade 4" OR "clade IV" OR "genotype 4" OR "clonal complex" OR genotype OR genotyping OR "diploid sequence type" OR "multilocus sequence typing" OR MLST OR "molecular typing" OR lineage OR "population structure" OR "genetic relatedness" OR clonal OR phylogenetic OR phylogenomic OR "whole genome sequencing" OR WGS OR "single nucleotide polymorphism" OR microsatellite OR "cross resistance" OR "zoonotic transmission" OR "environmental transmission" OR spillover OR resistance OR resistant OR susceptibility OR susceptible OR "azole tolerance" OR resistome OR "minimum inhibitory concentration" OR "epidemiological cutoff" OR ECOFF OR "non-wild-type" OR breakpoint OR breakpoints OR CLSI OR EUCAST OR "broth microdilution" OR "disk diffusion" OR fluconazole OR voriconazole OR itraconazole OR posaconazole OR isavuconazole OR ketoconazole OR clotrimazole OR miconazole OR "amphotericin B" OR echinocandin OR echinocandins OR anidulafungin OR caspofungin OR micafungin OR flucytosine OR "5-fluorocytosine" OR azole OR azoles OR triazole OR triazoles OR ERG11 OR ERG3 OR ERG5 OR Y132F OR S154F OR A395T OR C461T OR TAC1 OR MRR1 OR UPC2 OR CDR1 OR CDR2 OR MDR1 OR FKS1 OR FUR1 OR efflux OR overexpression OR aneuploidy OR isochromosome OR "loss of heterozygosity")

The public interface caps export at five hundred records per download and does not allow selection of a range of positions, so the master equation was retrieved in four segments by publication year: up to 2010, 408 records; 2011 to 2018, 401; 2019 to 2023, 417; and 2024 to 2027, 329. The last segment includes publication years after the current one because the platform

assigns anticipated year to articles with a future cover date. The four segments sum to 1,555, coinciding exactly with the yield of the line, with 1,481 unique digital object identifiers and no repetition between segments.

2.6 Lens.org

Interface: lens.org, academic account. Field prefixes confirmed on the platform: title, abstract, keyword and field of study. Scope factor: the same line yields 35,907 records across all fields and 15,709 restricted to the four prefixes above.

LEN-A. Population block. Yield: 15,709; Date: 10/09/2026

```
title:("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") OR abstract:("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") OR keyword:("Candida tropicalis" OR "C. tropicalis" OR "Candida paratropicalis") OR field_of_study:("Candida tropicalis")
```

The thematic block lines are not executable in this interface: it expands each block to four field indexes multiplied by the number of terms, generating strings that exhaust the server. The limitation is operational and is declared as such.

2.7 AGRIS

Interface: agris.fao.org. Anywhere field, with inclusion of bibliographic records from aggregators. Calibration: the quoted phrase search yields 1,801 records and the unquoted search 2,364, a difference of 563 corresponding to records in which the terms are not contiguous; the latter is adopted under the maximum sensitivity criterion and the difference is resolved in screening through EC1.

AGR-1. Population block. Yield: 2,364 reported, 2,365 retrieved; Date: 10/09/2026

```
Candida tropicalis
```

AGR-2. Agricultural compartment. Yield: 39; Date: 10/09/2026

```
Candida tropicalis AND (orchard OR fungicide OR tebuconazole OR difenoconazole OR postharvest OR rhizosphere OR "irrigation water")
```

AGR-3. Animal compartment. Yield: 121; Date: 10/09/2026

```
Candida tropicalis AND (veterinary OR poultry OR livestock OR swine OR cattle)
```

AGR-2 and AGR-3 are contained in full within AGR-1, with zero records outside it and a single record common to both; they document the contribution by compartment and are not imported separately.

2.8 Google Scholar

All lines are executed with the title operator, with the citation inclusion box ticked and without filters of date, document type or language. The platform limits navigation to one hundred pages of ten records, verified on the universal line. The figure declared in the header is an estimate of matches and not a count of navigable records; both figures are reported separately. The set entering screening is the deduplicated union of the five thematic lines; the universal line does not enter screening.

GS-1. Universal agent line, reference figure. Yield: 4,220 declared matches; Date: 12/09/2026

allintitle: "Candida tropicalis"

GS-2. Resistome. Yield: 106 declared, 97 retrieved; Date: 12/09/2026

allintitle: "Candida tropicalis" ("azole resistance" OR "antifungal resistance" OR "fluconazole resistance" OR ERG11)

GS-3. Agricultural compartment. Yield: 53 declared, 43 retrieved; Date: 12/09/2026

allintitle: "Candida tropicalis" (orchard OR agricultural OR fruit OR soil OR fungicide OR tebuconazole)

GS-4. Animal and environmental compartment. Yield: 86 declared, 76 retrieved; Date: 12/09/2026

allintitle: "Candida tropicalis" (veterinary OR animal OR environmental OR water OR wastewater OR seawater)

GS-5. Lineage and typing. Yield: 51 declared, 43 retrieved; Date: 12/09/2026

allintitle: "Candida tropicalis" ("clade 4" OR clade OR "multilocus sequence typing" OR MLST OR genotyping OR "diploid sequence type")

GS-6. Regional output in Spanish and Portuguese. Yield: 58 declared, 52 retrieved; Date: 12/09/2026

allintitle: "Candida tropicalis" (resistencia OR resistência OR antifúngica OR antifúngicos OR fluconazol OR sensibilidad OR sensibilidade)

Across the five thematic lines the platform declared 354 matches and 311 records were retrieved. Of these, 85 are entries the platform indexes only as a citation, without a page of their own, which include theses and conference papers and concentrate in the final pages of the relevance ordering.

2.9 Chinese-language sources

The strategy is decomposed into lines by block, since these platforms do not admit nesting of four blocks with simultaneous truncation. Both the simplified and the traditional forms of the species name are used, which are not interchangeable in the search engines. Automatic expansion functions, offered by default under the names cross-language search, synonym search and English-language extension, were expressly disabled in all three mainland platforms for incompatibility with reproducibility.

AIR-1. Airiti Library. Population block. Yield: 84 reported, 83 retrieved; Date: 11/09/2026

([ALL3]=("Candida tropicalis") OR [ALL3]=("熱帶念珠菌") OR [ALL3]=("C. tropicalis"))

Title, keyword and abstract field, precise search mode, without filters of date, language, country or document type. Calibration: the two full name forms yield 83 records and the abbreviated form adds one. Of the 83 retrieved, 31 are theses.

CNK-1. CNKI. Population block. Yield: 1,692; Date: 11/09/2026

(Title: 热带念珠菌(精确)) OR (Title: Candida tropicalis(精确))

Title field, precise mode. Calibration: the same query in the title, keyword and abstract scope yields 10,296 records, mostly literature on fungal infection in general, because the engine

decomposes the Chinese term into its constituent characters; restricted to the title field it yields 1,692. The two name forms joined by intersection yield 487 and joined by union 1,692. Composition: 1,584 journal articles, 84 theses and dissertations, 24 conference papers.

WAN-1. Wanfang Data. Population block. Yield: 1,609; Date: 11/09/2026

题名:热带念珠菌 OR 题名:Candida tropicalis

Title field, precise mode. Calibration: with the English-language extension active and a three-category document type filter the line yields 1,410 records; with the extension disabled and the same filter, 1,390; without the type filter, 1,609. Not retrieved: export requires registration with a mainland China mobile number, which the reviewing team does not hold. Declared as an access restriction.

VIP-1. Chongqing VIP Information. Population block. Yield: 2,601, of which 159 in Chinese; Date: 11/09/2026

篇名:热带念珠菌 OR 篇名:Candida tropicalis

Title field, precise mode. Calibration: with both automatic expansions active the line yields 3,040 records, of which 587 in Chinese, so the expansion adds 439 records. The Chinese-language subset is retrieved as the exclusive contribution of the source, since the 2,442 foreign-language records correspond to international literature already retrieved by the multidisciplinary sources of the strategy.

2.10 Regional open-access sources

A single population line is executed in each, since the volume does not justify decomposition into blocks. The scope is the broadest each platform admits without resorting to full text.

SCI-1. SciELO. Population block. Yield: 118; Date: 11/09/2026

"Candida tropicalis"

All indexes, without filters. Calibration: the title field yields 30 records. Composition by language: 69 in English, 33 in Spanish, 16 in Portuguese. The export of the platform omits the abstract field.

RED-1. Redalyc. Population block. Yield: 201; Date: 11/09/2026

"Candida tropicalis"

Without filters of year, language, discipline or country. Composition by country: Colombia 53, Brazil 34, Mexico 28, Cuba 21, Venezuela 20. By language: 129 in Spanish, 58 in English, 13 in Portuguese. By discipline: 89 medicine, 30 biology, 20 agricultural sciences.

AJO-1. African Journals Online. Population block. Yield: 85; Date: 12/09/2026

"Candida tropicalis"

Without filters of date, journal or document type. Digital object identifier available in 58 of the 85 records.

KOR-1. KoreaMed. Population block. Yield: 28; Date: 12/09/2026

Candida tropicalis

All fields scope, without filters. Range 1995 to 2024.

JST-1. J-Stage. Population block. Yield: 326; Date: 12/09/2026

Article title: Candida tropicalis OR Abstract: Candida tropicalis OR Keyword(s): Candida tropicalis

Three rows of the advanced search joined by the union operator, since the platform offers no combined title, abstract and keyword scope. Automatic Japanese to English translation disabled. Calibration: the query in full text without quotation marks yields 21,895 records and with quotation marks 1,117, so the scope restriction reduces retrieval by a factor of sixty-seven. Range 1953 to 2026.

2.11 Preprint servers

PRE-1a. bioRxiv and medRxiv. Population block. Yield: 17; Date: 11/09/2026

Abstract or Title: Candida tropicalis

Phrase match mode, both servers included in a single query, without filters. Sixteen records come from bioRxiv and one from medRxiv.

PRE-1b. Research Square. Population block. Yield: 309 reported, 301 retrieved; Date: 11/09/2026

Title: Candida tropicalis

Publication status unfiltered. The engine does not support exact phrase search: the general query without field restriction yields 931 records and the title-restricted query 309, with and without quotation marks. Of the 301 retrieved, 27 name the species in the title and 16 of those have no published journal version. Breakdown by status: 144 posted, 95 published in journal, 58 under review, 4 in revision.

PRE-1c. Preprints.org. Population block. Yield: 7; Date: 11/09/2026

Candida tropicalis

Exact match type, without filters of period, subject area or manuscript type. Calibration: partial match yields 107 records.

PRE-1d. SSRN. Population block. Yield: 62; Date: 11/09/2026

Candida tropicalis

Title, abstract and keywords scope, with boolean search enabled and without filters of date or author. Calibration: the scope restricted to title alone yields 13 records, so abstract and keywords add 49.

3. SUMMARY OF SOURCES, LINES OF RECORD AND YIELDS

Table 1. Sources executed, lines of record and yields

Source	Line of record for screening	Records	Date
PubMed	PM-A population; PM-M as screening priority	7,028; 1,194	07/09/2026
Scopus	SC-A population; SC-M as screening priority	14,500; 2,715	08/09/2026
Lens.org	LEN-A population	15,709	10/09/2026
AGRIS	AGR-1 population	2,364 reported; 2,365 retrieved	10/09/2026
LILACS Plus	LIL-2 population by all words	1,857 returned; 1,668 unique	10/09/2026
CNKI	CNK-1 population, title field	1,692	11/09/2026
Dimensions	DIM-M master equation, by export limit	1,555	10/09/2026
ScienceDirect	Fifteen sweeps, no master equation	1,149 non-additive sum	09/09/2026
J-Stage	JST-1 population	326	12/09/2026
Google Scholar	Union of GS-2 to GS-6	354 declared; 311 retrieved	12/09/2026
Research Square	PRE-1b population, title field	309 reported; 301 retrieved	11/09/2026
Redalyc	RED-1 population	201	11/09/2026
VIP	VIP-1, Chinese-language subset	2,601; 159 in Chinese	11/09/2026
SciELO	SCI-1 population, all indexes	118	11/09/2026
African Journals Online	AJO-1 population	85	12/09/2026
Airiti Library	AIR-1 population	84 reported; 83 retrieved	11/09/2026
SSRN	PRE-1d population	62	11/09/2026
KoreaMed	KOR-1 population, all fields	28	12/09/2026
bioRxiv and medRxiv	PRE-1a population, joint query	17	11/09/2026
Preprints.org	PRE-1c population, exact match	7	11/09/2026
Wanfang Data	WAN-1. Executed, not retrieved	1,609	11/09/2026
Web of Science, Core Collection	Not executed. No subscription	Not applicable	30/08/2026
Embase	Not executed. No subscription	Not applicable	30/08/2026
AGRICOLA	Not executed. Its provider contributes 833 records to AGR-1	Not applicable	10/09/2026
CAB Abstracts and Global Health	Not executed. Platform not accessible	Not applicable	12/09/2026

4. RETRIEVAL PROCEDURE

PubMed, Scopus, Lens.org, LILACS, SciELO, CNKI and ScienceDirect allow download in reference manager format, and the sets were obtained by that route. Dimensions imposes a five-hundred-record limit per download. Wanfang Data conditions export on registration requiring a mainland China mobile number, and J-Stage on account creation. AGRIS, Redalyc, Research Square, African Journals Online, KoreaMed and Google Scholar do not offer it, or limit it to citation styles not importable to a reference manager.

In these last six, retrieval was performed by extracting the metadata each platform presents on its own results pages, using tools developed for this review. The procedure reproduces navigation through results pages and transcribes the visible fields into an importable format, accessing no content other than that displayed to the user and circumventing no access control. In each source the number of extracted records was checked against the yield the platform reports and identifier uniqueness was verified. Residual differences are recorded in the note of each source: one record in excess in AGRIS, attributable to recalculation of the total during pagination; eight fewer in Research Square, from repetition of records between pages; and one fewer in Airiti Library, from a reproducible failure of its export function.

Terms of use of every source retrieved by this route are verified before search closure and the outcome recorded. This verification bears in particular on Google Scholar, whose terms of service restrict automated access.

The tools employed, the export files and the verification records are deposited in the open-access repository of the project together with the analysis code.

5. ROUTES OTHER THAN DATABASES

Reference list tracking and forward citation searching are executed once screening is closed, limited to a single iteration, and applied exhaustively to included studies from non-clinical compartments and to identified prior syntheses. Forward citation searching is run in Scopus, Google Scholar and Lens.org. Results are counted separately in the flow diagram. Thesis repositories are searched after screening closure. Contact with authors for study identification and conference proceedings were discarded a priori.

PubMLST and isolate metadata deposited in NCBI BioSample and the Sequence Read Archive constitute isolate-level data and not peer-reviewed studies; their treatment is that of an auxiliary triangulation analysis, reported separately and never merged with pooled proportions.

6. RELATIVE RECALL CHECK

Before closing identification, each source is checked for recovery of a seed set of known and necessarily eligible studies: the multilocus typing work defining the clade scheme of the species, the two Taiwanese orchard surveys of 2012 and 2018, the study of molecular mechanisms of azole resistance in invasive isolates from China, and the review of the species prepared for the fungal priority pathogens list, the last as a recovery control although its design is excluded. The proportion of the seed recovered by each source is recorded.